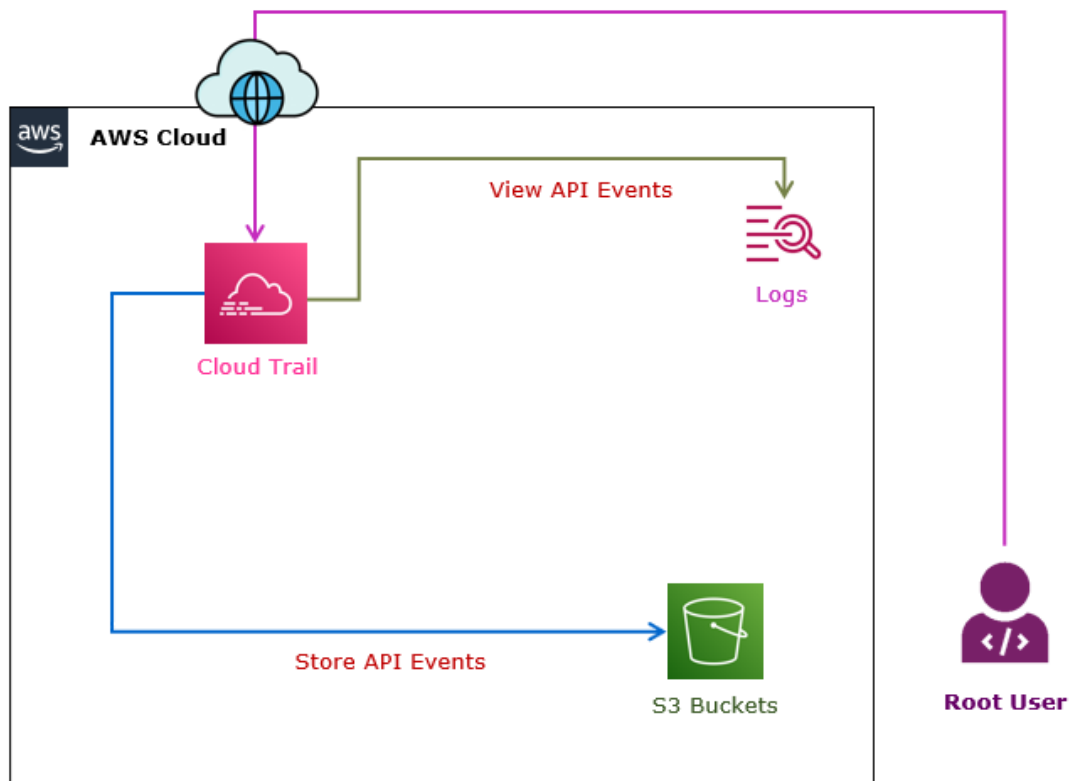


Optional Task: If time permits

Task 1: Store API Calls in S3 Bucket

In this task, you will store the trails into S3 bucket for compliance purpose and integrate the trail with CloudWatch to generate the alerts.



Step 1: Create AWS S3 buckets

1. In the **AWS Management Console**, on the **Services** menu, click **S3**.
2. Click **Create bucket**.
 - a. **Bucket name**: Write **cloudtrail-123**.

Note: Replace **123** with a random number to make bucket name unique.

- b. **Region**: Dropdown and select **US East (N. Virginia)**.

Note: Leave other details as default.

- c. Click **Create bucket**.

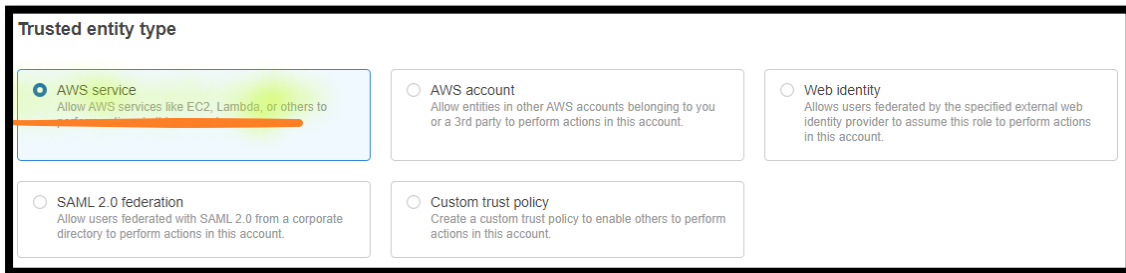
Step 2: Create IAM Roles for CloudTrail Logs

3. In the **AWS Management Console**, on the **Services** menu, click **IAM**.

4. Select **Roles**, click on **Create role**.

a. In the **Select trusted entity** section:

i. **Trusted entity type**: Select **AWS service**.



Trusted entity type

☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to

☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

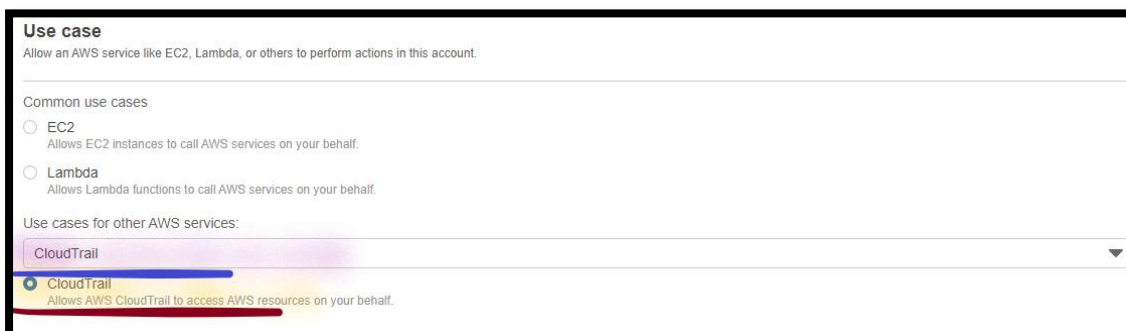
☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

ii. **Common use cases**: Dropdown and select **CloudTrail**.

a) Select **CloudTrail**.

iii. Select **Next**.



Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Common use cases

☐ **EC2**
Allows EC2 instances to call AWS services on your behalf.

☐ **Lambda**
Allows Lambda functions to call AWS services on your behalf.

Use cases for other AWS services:

CloudTrail

☒ **CloudTrail**
Allows AWS CloudTrail to access AWS resources on your behalf.

b. In the **Add Permissions** section:

Note: Leave all the details as default.

i. Select **Next**.

c. In the **Name, review, and create** section:

Note: Leave all the details as default.

i. Click **Create role**.

Note: **Wait**, till Role **AWSServiceRoleForCloudTrail** gets created.

Step 3: Enabling CloudTrail event logging for S3 buckets

5. In the **AWS Management Console**, on the **Services** menu, click **CloudTrail**.
6. Dropdown and select **US East (N. Virginia)** region.
7. Click **Trails**.
 - a. Select **Create trail**.
 - i. In the **General details** section:
 - a) **Trail name:** Write **cloudtrail-events**.
 - b) **Storage location:** Select **Use existing S3 bucket**.
 - c) **Trail log bucket name:** Select **Browse**.
 - 1) Select **cloudtrailevents-123** bucket.
 - 2) Select **Choose**.
 - d) **Prefix - optional:** Write **events**.

The screenshot shows the 'Create trail' page in the AWS Management Console, specifically the 'General details' section. The 'Trail name' field is filled with 'cloudtrail events'. The 'Storage location' section has 'Use existing S3 bucket' selected. The 'Trail log bucket name' field is filled with 'cloudtrailevents-123', and the 'Browse' button is highlighted. The 'Prefix - optional' field is filled with 'events'. At the bottom, a note states: 'Logs will be stored in cloudtrailevents-123/events/AWSLogs/504340965905'.

Trail name
Enter a display name for your trail.
cloudtrail events
3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

☐ Enable for all accounts in my organization
To review accounts in your organization, open AWS Organizations. [See all accounts](#)

Storage location [Info](#)

☐ Create new S3 bucket
Create a bucket to store logs for the trail.

☒ Use existing S3 bucket
Choose an existing bucket to store logs for this trail.

Trail log bucket name
Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.
cloudtrailevents-123 [Browse](#)

Prefix - optional
events
Logs will be stored in cloudtrailevents-123/events/AWSLogs/504340965905

- e) **Log file SSE-KMS encryption:** **Uncheck** the **Checkmark** against **Enabled**.
- f) **Log file validation:** **Uncheck** the **Checkmark** against **Enabled**.



Log file SSE-KMS encryption [Info](#)

☐ Enabled

▼ Additional settings

Log file validation [Info](#)

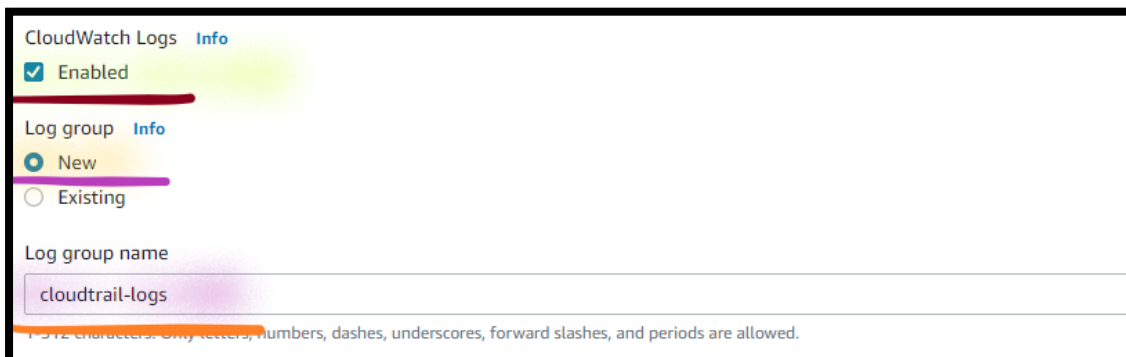
☐ Enabled

SNS notification delivery [Info](#)

☒ Enabled

ii. In the **CloudWatch Logs - optional** section:

- a) **CloudWatch Logs:** **Enable** the **Checkmark** against **Enabled**.
 - 1) **Log group:** Select **New**.
 - 2) **Log group name:** Write **cloudtrail-logs**.



CloudWatch Logs [Info](#)

☒ Enabled

Log group [Info](#)

☒ New

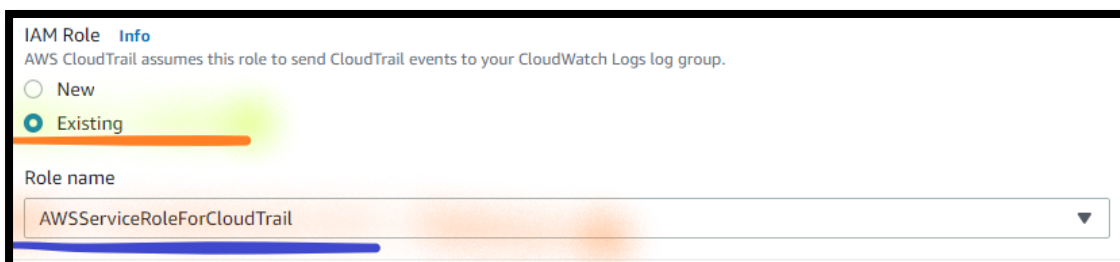
☐ Existing

Log group name

cloudtrail-logs

1-512 characters. Only letters, numbers, dashes, underscores, forward slashes, and periods are allowed.

- b) **IAM Role:** Select **Existing**.
 - 1) **Role name:** Dropdown and select **AWSServiceRoleForCloudTrail**.



IAM Role [Info](#)

AWS CloudTrail assumes this role to send CloudTrail events to your CloudWatch Logs log group.

☐ New

☒ Existing

Role name

AWSServiceRoleForCloudTrail ▼

- c) Select **Next**.

iii. In the **Events** section:

Note: Leave all the details as default.

a) Select **Next**.

iv. In the **Management Events** section:

Note: Leave all the details as default.

a) Select **Next**.

v. In the **Review and create** section:

a) Select **Create trail**.

Step 4: View the events from S3 buckets

8. In the **AWS Management Console**, on the **Services** menu, click **S3**.

9. Select **Bucket**.

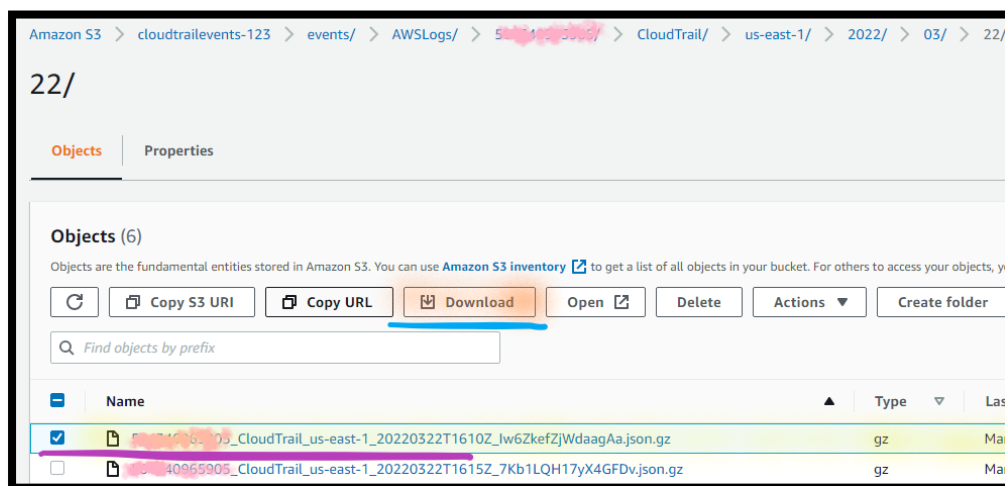
a. Open **cloudtrailevents-123**.

b. Open **events**.

c. Open the **folders** and **subfolders** until you can see the **Object**.

Note: You can see the Object name as "**account-name_cloudtrail_region_xxx**".

d. Select the **object** and **Download**.



- e. Open the **deliverystream-s3-xxxx** file in the **Notepad**.

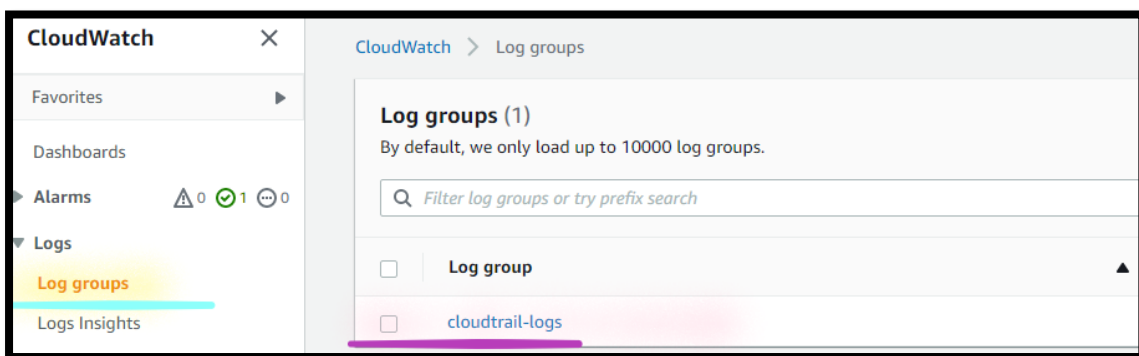
Note: You can see the **API details**.

Step 5: Monitor the CloudWatch Events

10. In the **AWS Management Console**, on the **Services** menu, click **CloudWatch**.

11. **Expand** the **Logs**.

- a. Click on the **Log groups**.

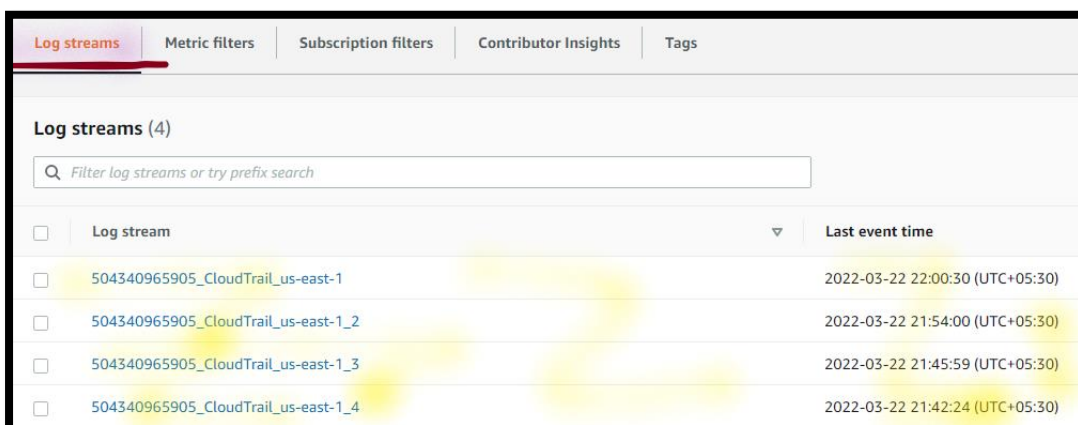


- b. Open the **cloudtrail-logs** log group.

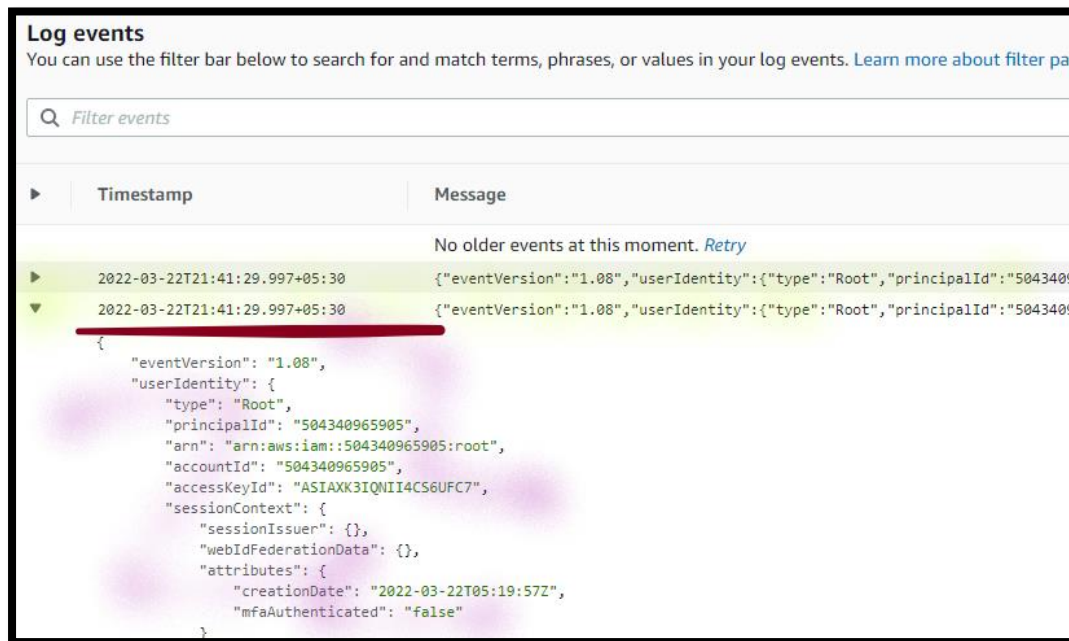
Note: You can see the **Log streams** generated by **Cloud Trail** with the **Date & Time**.

- c. Select the **Log streams**.

- i. Open the **latest Log streams**.



Note: **Inspect the contents** of all the recent **entries** of all the log streams.



Task 2: Cleanup the Environment

Step 1: Delete Trail

12. In the **AWS Management Console**, on the **Services** menu, click **CloudTrail**.

13. Click **Trails**.

a. Select **cloudtrail-events**.

i. Select **Delete**.

ii. Select **Delete**.

Step 2: Delete Log group

14. In the **AWS Management Console**, on the **Services** menu, click **CloudWatch**.

15. **Expand** the **Logs**.

a. Click on the **Log groups**.

b. Select the **cloudtrail-logs**.

i. Select **Actions**.

- ii. Select **Delete log groups(s)**.
- iii. Select **Delete**.

Step 3: Delete the S3 Bucket Object

16. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

- a. Select **cloudtrailevents-123** bucket.
 - i. Select **Empty**.
 - a) **Type permanently delete** to **Delete all the objects**.
 - b) Select **Empty**.
 - c) Select **Exit**.
 - b. Select **cloudtrailevents-123** bucket.
 - i. Select **Delete**.
 - d) **Type cloudtrailevents-123** to **Delete the bucket**.
 - e) Select **Delete**.